relativity and quantum mechanics, has allowed us now to plant these possibilities firmly at the forefront of cutting-edge science. This success emboldens us to ask whether any of the other mysterious properties of our universe that have stubbornly resisted resolution for decades might also succumb to the powers of string theory. Foremost among these is the notion of black hole entropy. This is the arena in which string theory has most impressively flexed its muscles, successfully solving a quarter-century-old problem of profound significance.

Entropy is a measure of disorder or randomness. For instance, if your desk is cluttered high with layer upon layer of open books, half-read articles, old newspapers, and junk mail, it is in a state of high disorder, or high entropy. On the other hand, if it is fully organized with articles in alphabetized folders, newspapers neatly stacked in chronological order, books arranged in alphabetical order by author, and pens placed in their designated holders, your desk is in state of high order or, equivalently, low entropy. This example illustrates the essential idea, but physicists have given a fully quantitative definition to entropy that allows one to describe something's entropy by using a definite numerical value: Larger numbers mean greater entropy, smaller numbers mean less entropy. Although the details are a little complicated, this number, roughly speaking, counts the possible rearrangements of the ingredients in a given physical system that leave its overall appearance intact. When your desk is neat and clean, almost any rearrangement—changing the order of the newspapers, books, or articles, moving the pens from their holders—will disturb its highly ordered organization. This accounts for its having low entropy. On the contrary, when your desk is a mess, numerous rearrangements of the newspapers, articles, and junk mail will leave it a mess and therefore will not disturb its overall appearance. This accounts for its having high entropy.

Of course, a description of rearranging books, articles, and newspapers on a desktop—and deciding which rearrangements "leave its overall appearance intact"—lacks scientific precision. The rigorous definition of entropy actually involves counting or calculating the number of possible rearrangements of the microscopic quantum-mechanical properties of the elementary constituents of a physical